

WIRELESS SENSOR LOGGER WITH OFFLINE FAILOVER BUFFERING

Abstract

This project implements a wireless sensor logging system using two ESP32 boards. The transmitter reads an analog sensor and sends data to the receiver using ESP-NOW. The receiver uploads validated readings to ThingSpeak for real-time monitoring. If communication fails, the transmitter stores readings in SPIFFS flash memory and automatically retransmits them when the connection is restored, preventing data loss.

Objectives

- Enable ESP-NOW communication between two ESP32 boards.
- Log analog sensor data reliably.
- Buffer failed transmissions in SPIFFS.
- Automatically resend buffered data.
- Display sensor readings on ThingSpeak.

Working Principle

The transmitter samples an LDR or potentiometer every 5 seconds and sends the data with sequence information. The receiver acknowledges successful reception and forwards the readings to ThingSpeak over Wi-Fi. If acknowledgment is not received, the transmitter stores the reading locally in SPIFFS. Once communication is restored, stored readings are transmitted first before new data, maintaining the correct order.

Hardware & Software

Hardware: 2× ESP32 boards, Potentiometer, breadboard, jumper wires, USB cables.

Software: Arduino IDE, ESP-NOW library, SPIFFS, HTTPClient, ThingSpeak.

Results

The system successfully provides reliable wireless sensor monitoring. Data is uploaded to the cloud during normal operation, while temporary communication failures do not cause data loss because readings are buffered and later synchronized automatically.

Applications

Industrial monitoring, environmental sensing, smart agriculture, remote IoT telemetry, and educational embedded systems projects.

Conclusion

The project demonstrates a low-cost and reliable IoT telemetry system by combining ESP-NOW communication, cloud monitoring, and flash-based offline buffering. It ensures continuous data logging even during temporary network failures, making it suitable for practical embedded and industrial applications.